



General Details

Assessor	Samuel Prieto Ayllon
Assessment Date	03/07/2025
Assigned Reviewer	Samuel Prieto Ayllon
Next Review Date	03/07/2026

Signed Off By	Oraib AlKheetanRojas
Signed Off On	19/05/2026
Sign Off Comments	

Operation Assessed	Subsea Manipulator – Reach Alpha Operation
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Associated with specific area

 Basement 2

 B2 029 (Robotics Lab)

 CTP

Description of work area and/or activity assessed

The Reach Alpha is a compact, electrically driven subsea manipulator (5-function variant) used for precision grasping, cutting, and inspection tasks on remotely operated or bench-top test platforms. Typical activities include integration, maintenance, and tele-operated motion within the lab or on small ROVs in test tanks. Operations involve low-voltage DC power, serial communications, and occasional jaw changes.

Reason for Risk Assessment

To document and control any health, safety, and equipment-damage risks introduced by actively using the Reach Alpha manipulator in research and integration activities.

Overall Current Risk

Moderate Risk



Average Number of Persons Affected

2  Employees

2  Students

2  Visitors

Current Rating	Hazard Information	Task Status
Moderate Risk 	Moving Parts The manipulator's five rotary joints and electric grabber can move rapidly and forcefully. Uncontrolled motion can strike operators or collide with fixtures, leading to crush injuries or equipment damage.	0 0 0
	Measures Currently in place to prevent risk of injury	
	• Install and operate below head height. • Avoid physical interaction with the end-effector or limbs to prevent crush and/or cut hazards • Use Reach Control virtual obstacles and speed/torque limits to restrict workspace. • Keep untrained personnel behind barriers and signage during motion. • Perform bench tests at reduced speed before full-power deployment.	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Moderate Risk 	Pinch Points Pinch/crush zones exist between the jaws, at each limb intersection, and in cutter accessories. Fingers or loose clothing can be trapped when joints close or the grabber actuates.	0 0 0
	Measures Currently in place to prevent risk of injury	
	• Wear safety goggles and avoid hand contact with end-effector and joints. • Ensure work area is clear before energising motors. • Train operators to stow the arm and power off before tool change. • Use software "Pause" or "Enable" buttons to keep motors disabled until the workspace is verified clear.	

Potential Rating	Additional Controls Required
Moderate Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Low Risk 	Electricity The system operates on 18 – 30 V DC (up to 35 W). Damaged cables, incorrect grounding, or exposed conductors can cause mild electric shock, short circuits, or equipment failure.	0 0 0
	Measures Currently in place to prevent risk of injury	
	• Power supplied via Safety Extra-Low Voltage (SELV). • Interface cable and bulkhead connectors are sealed; connectors visually inspected before use. • Serial comms grounding scheme followed to avoid ground loops. • Cables strain-relieved and routed away from sharp edges. • Only trained personnel permitted to modify wiring.	

Potential Rating	Additional Controls Required
Low Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Low Risk 	Sharp Objects When fitted with razor-style cutter jaws, the end-effector can sever cables, hoses, or skin on contact. Accidental activation or unexpected motion can cause serious lacerations to hands or nearby operators. The manual highlights jaws as a combined crush/cut hazard that requires the workspace to be clear before motion.	0 0 0
	Measures Currently in place to prevent risk of injury	
	• Fit protective jaw covers when the arm is idle or during transport. • Issue cut-resistant gloves (EN 388 level C or higher) for any task requiring proximity to the jaws. • Program Reach Control speed and force limits specifically for cutting operations; disable high-force modes during bench testing. • Establish a “no-hands” rule: only use tools (e.g., pliers) to load or remove materials being cut. • Keep the manipulator stowed and power-off before jaw changes; recalibrate (“zero”) jaws after each change as required by the manual to avoid uncommanded motion.	

Potential Rating	Additional Controls Required
Low Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.